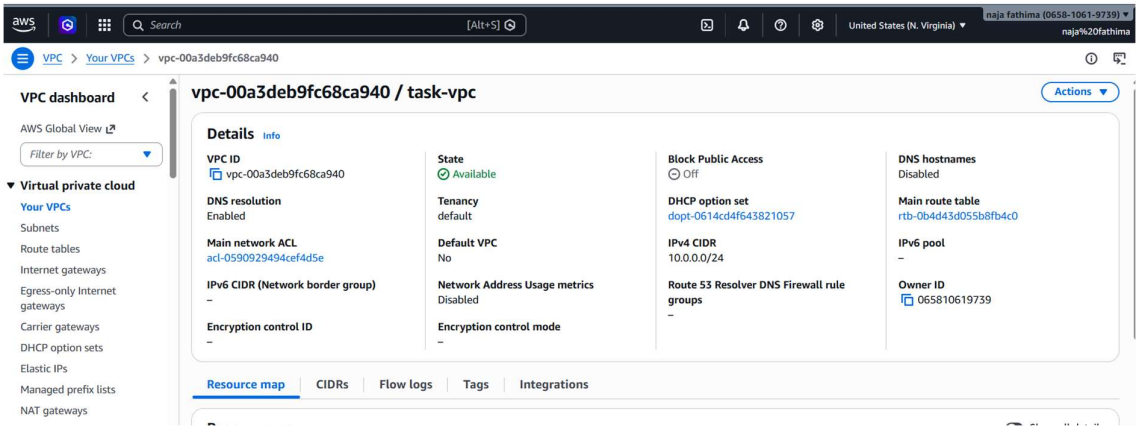
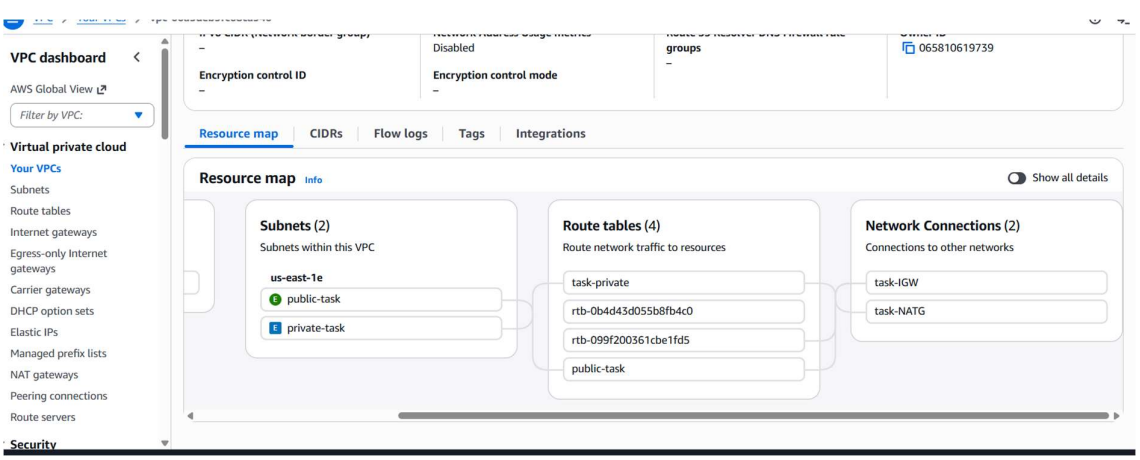
Implement NAT Gateway Setup

Description

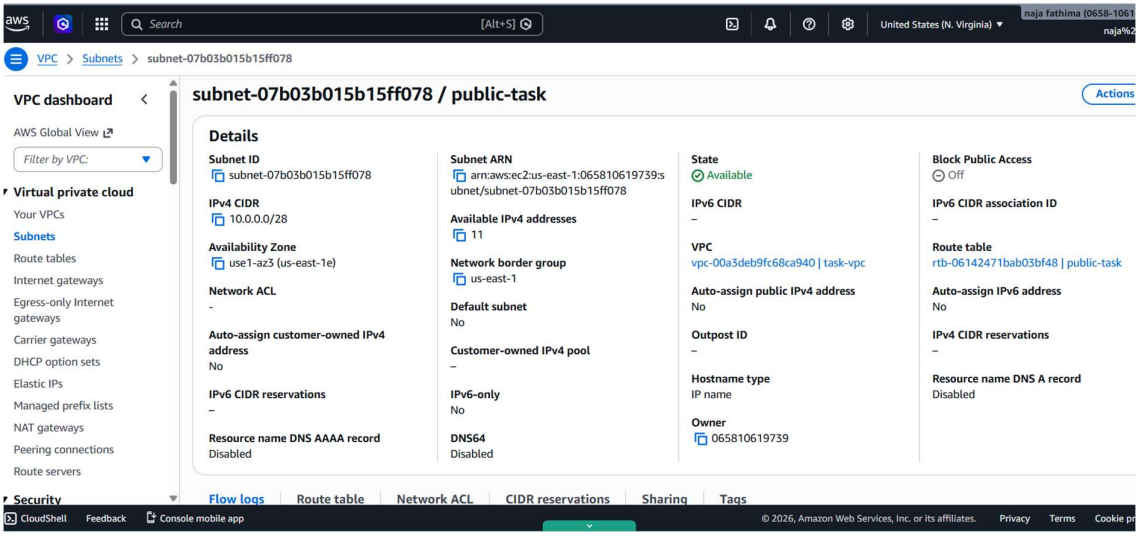
i. Create two instances, one in a public subnet and one in a private subnet
ii. Host a website inside public web server
iii. Host a database inside private database server
iv. Connect to DB server only from its webserver, shouldn't have a public access to server even if the security group has open entries
v. DB Server should have internet access. Try to update the software packages, patching, etc

Step 1: Created VPC and 2 subnets

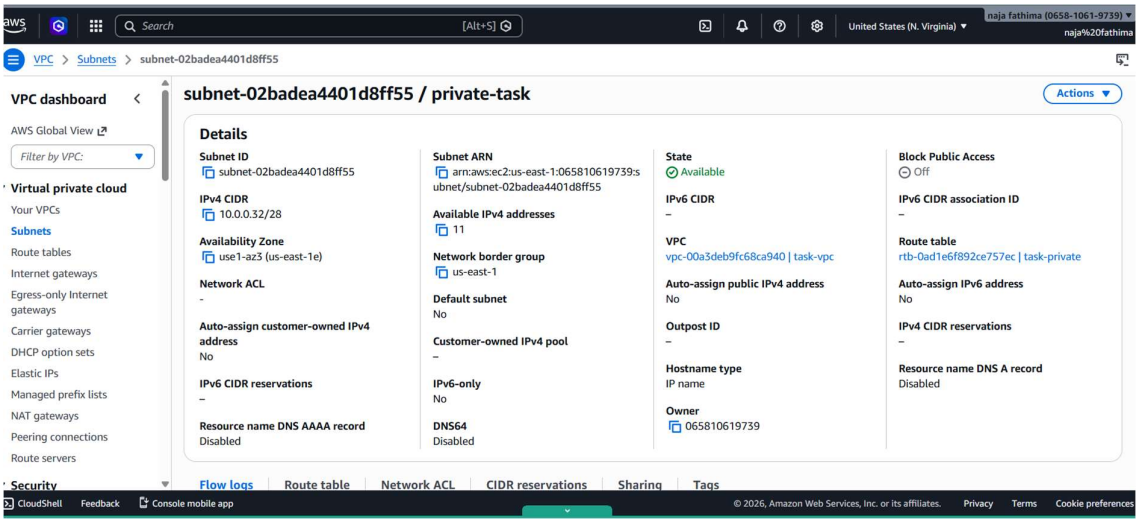




Public subnet

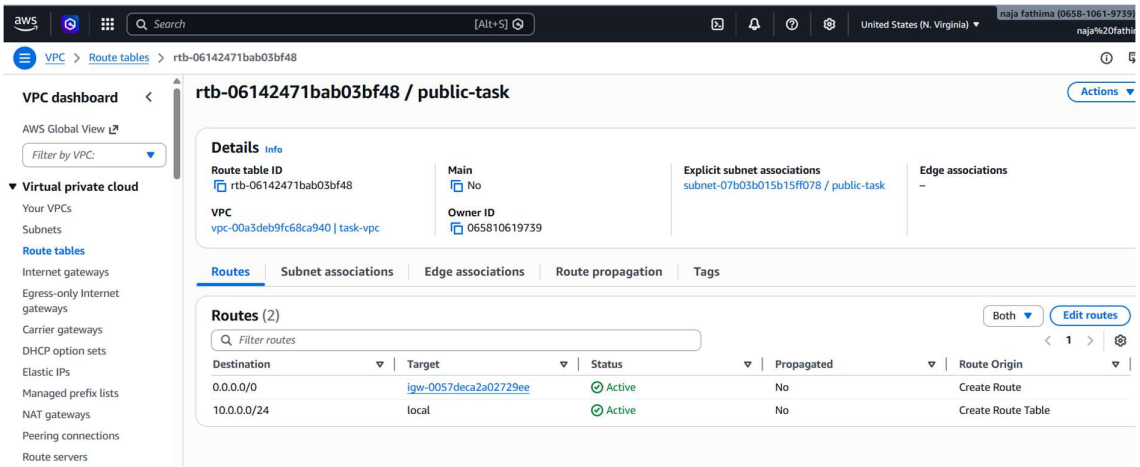


Private subnet

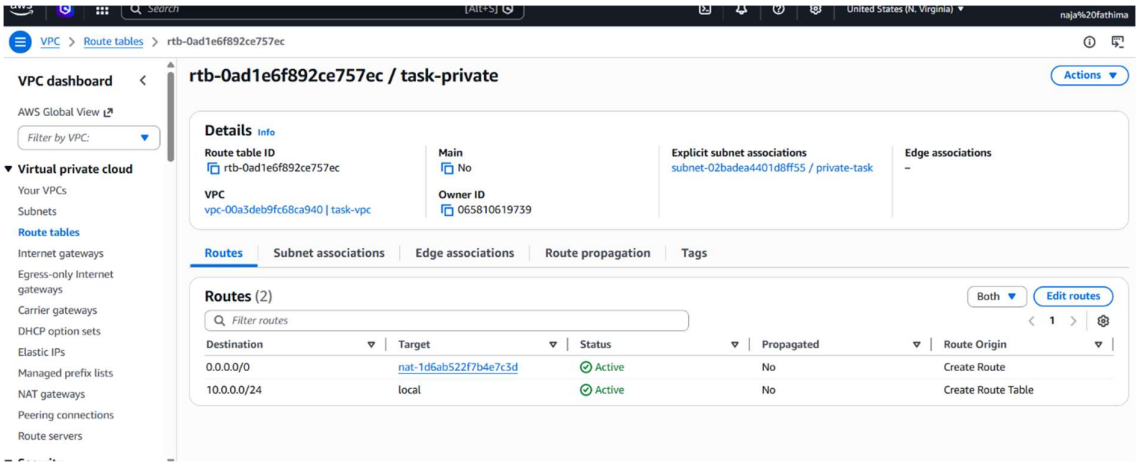


Step 2: Created 2 route tables, Internet gateway and NAT gateway

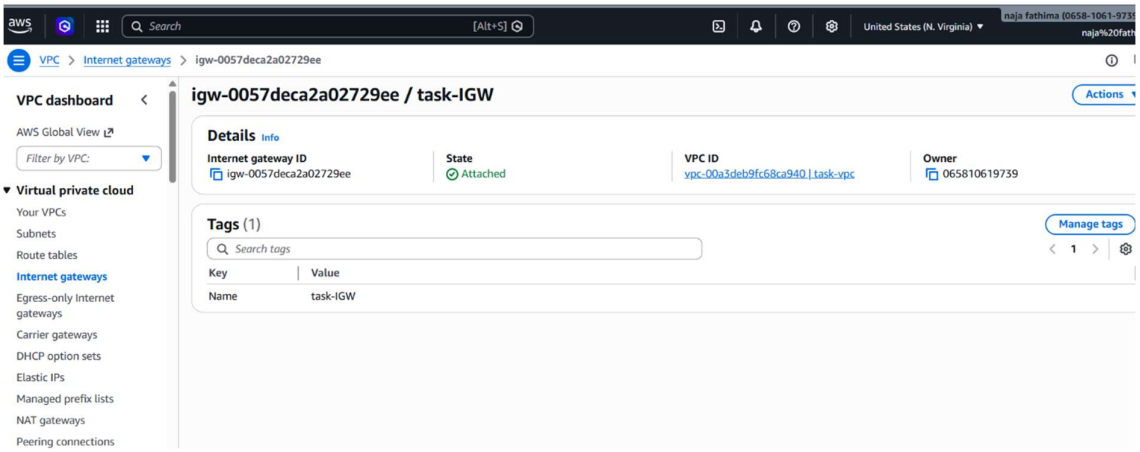
Public-route-table



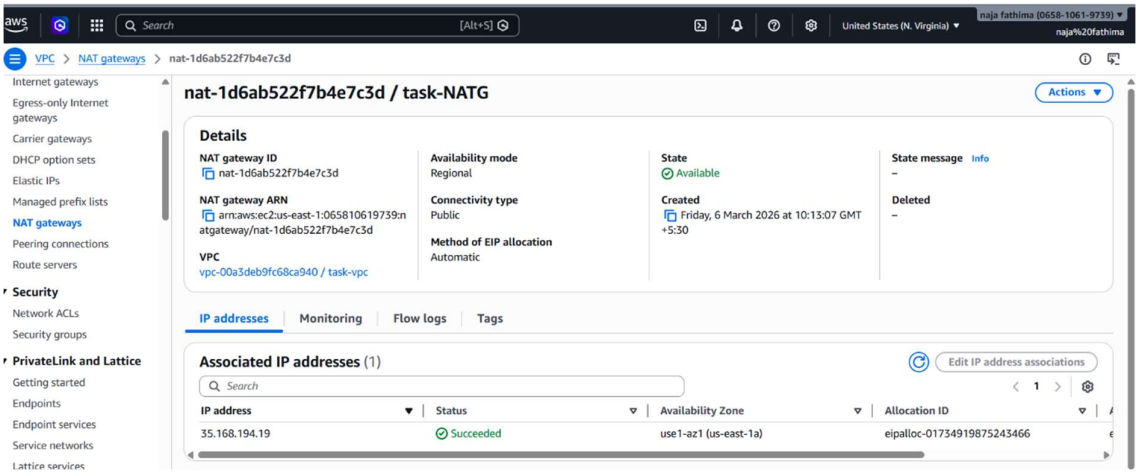
Private-route-table



Internet gateway



NAT gateway



Step 3: Launched public subnet and private subnet

Connected private subnet from public subnet

```
$ ssh -i "new-project.pem" ec2-user@54.165.82.13
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Fri Mar  6 04:58:52 2026 from 49.47.197.19
[ec2-user@ip-10-0-0-14 ~]$ ssh -i new-project.pem ec2-user@10.0.0.41
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
[ec2-user@ip-10-0-0-41 ~]$ top
```

Step 4: Installed httpd, WordPress packages on public server

```
[ec2-user@ip-10-0-0-14 ~]$ sudo dnf install httpd -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "
subscription-manager" to register.

Red Hat Enterprise Linux 9 for x86_64 - AppStre  66 kB/s | 4.5 kB     00:00
Red Hat Enterprise Linux 9 for x86_64 - BaseOS   57 kB/s | 4.1 kB     00:00
Red Hat Enterprise Linux 9 Client Configuration  27 kB/s | 1.5 kB     00:00
Dependencies resolved.
================================================================================
 Package            Arch   Version              Repository                  Size
================================================================================
Installing:
 httpd              x86_64 2.4.62-7.el9_7.3     rhel-9-appstream-rhui-rpms  47 k
Installing dependencies:
 apr                x86_64 1.7.0-12.el9_3       rhel-9-appstream-rhui-rpms 126 k
 apr-util           x86_64 1.6.1-23.el9         rhel-9-appstream-rhui-rpms  97 k
 apr-util-bdb       x86_64 1.6.1-23.el9         rhel-9-appstream-rhui-rpms  14 k
 httpd-core         x86_64 2.4.62-7.el9_7.3     rhel-9-appstream-rhui-rpms 1.5 M
 httpd-filesystem   noarch 2.4.62-7.el9_7.3     rhel-9-appstream-rhui-rpms  12 k
 httpd-tools        x86_64 2.4.62-7.el9_7.3     rhel-9-appstream-rhui-rpms  81 k
 mailcap            noarch 2.1.49-5.el9         rhel-9-baseos-rhui-rpms     35 k
 redhat-logos-httpd noarch 90.5-1.el9_6.1       rhel-9-appstream-rhui-rpms  16 k
Installing weak dependencies:
 apr-util-openssl   x86_64 1.6.1-23.el9         rhel-9-appstream-rhui-rpms  17 k
 mod_http2          x86_64 2.0.26-5.el9         rhel-9-appstream-rhui-rpms 163 k
 mod_lua            x86_64 2.4.62-7.el9_7.3     rhel-9-appstream-rhui-rpms  59 k

Transaction Summary
================================================================================
```

```
[ec2-user@ip-10-0-0-14 ~]$ sudo dnf install php php-mysqlnd php-fpm php-json php-gd php-mbstring php-xml php-curl -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 0:02:00 ago on Fri 06 Mar 2026 05:14:31 AM UTC.
Dependencies resolved.
================================================================================================================================
 Package                                    Architecture                        Version                                      Rep
================================================================================================================================
Installing:
 php                                        x86_64                              8.0.30-5.el9_7                               rhe
 php-common                                 x86_64                              8.0.30-5.el9_7                               rhe
 php-fpm                                    x86_64                              8.0.30-5.el9_7                               rhe
 php-gd                                     x86_64                              8.0.30-5.el9_7                               rhe
 php-mbstring                               x86_64                              8.0.30-5.el9_7                               rhe
 php-mysqlnd                                x86_64                              8.0.30-5.el9_7                               rhe
 php-xml                                    x86_64                              8.0.30-5.el9_7                               rhe
Installing dependencies:
 fontconfig                                 x86_64                              2.14.0-2.el9_1                               rhe
 gd                                         x86_64                              2.3.2-3.el9                                  rhe
 libjpeg-turbo                              x86_64                              2.1.3-2.el9                                  rhe
```

```
[ec2-user@ip-10-0-0-14 html]$ sudo wget https://wordpress.org/latest.tar.gz
--2026-03-06 05:18:35--  https://wordpress.org/latest.tar.gz
Resolving wordpress.org (wordpress.org)... 198.143.164.252, 2607:f978:5:8002::c68f:a4fc
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27062929 (26M) [application/octet-stream]
Saving to: 'latest.tar.gz'

latest.tar.gz                                       100%[===================================================================================================>]  2

2026-03-06 05:18:36 (46.2 MB/s) - 'latest.tar.gz' saved [27062929/27062929]

[ec2-user@ip-10-0-0-14 html]$ sudo tar -xzf latest.tar.gz
[ec2-user@ip-10-0-0-14 html]$ ls
latest.tar.gz  wordpress
[ec2-user@ip-10-0-0-14 html]$ sudo copy -R wordpress/* /var/www/html/
sudo: copy: command not found
[ec2-user@ip-10-0-0-14 html]$ sudo cp -R wordpress/* /var/www/html/
[ec2-user@ip-10-0-0-14 html]$ ls
index.php  license.txt  wordpress  wp-admin  wp-comments-post.php  wp-content  wp-includes  wp-load.php  wp-mail.php  wp-signup.php  xmlrpc.php
latest.tar.gz  readme.html  wp-activate.php  wp-blog-header.php  wp-config-sample.php  wp-cron.php  wp-links-opml.php  wp-login.php  wp-settings.php  wp-trackback.php
[ec2-user@ip-10-0-0-14 html]$ sudo chown -R apache:apache /var/www/html
```

```
[ec2-user@ip-10-0-0-14 html]$ sudo cp /var/www/html/wp-config-sample.php /var/www/html/wp-config.php
[ec2-user@ip-10-0-0-14 html]$ sudo systemctl restart httpd
```

Step 5: Installed and configured database on private subnet

```
[ec2-user@ip-10-0-0-41 ~]$ sudo dnf install mariadb-server -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 0:04:17 ago on Fri 06 Mar 2026 05:22:04 AM UTC.
Dependencies resolved.
================================================================================================================================
 Package                                    Architecture                    Version                                    Repositor
================================================================================================================================
Installing:
 mariadb-server                             x86_64                          3:10.5.29-3.el9_7                          rhel-9-ap
Installing dependencies:
 libaio                                     x86_64                          0.3.111-13.el9                             rhel-9-ba
 mariadb-errmsg                             x86_64                          3:10.5.29-3.el9_7                          rhel-9-ap
 mysql-selinux                              noarch                          1.0.14-1.el9_6                             rhel-9-ap
 perl-DBD-MariaDB                           x86_64                          1.21-17.el9                                rhel-9-ap
 perl-DBI                                   x86_64                          1.643-9.el9                                rhel-9-ap
 perl-DynaLoader                            x86_64                          1.47-481.1.el9_6                           rhel-9-ap
 perl-File-Copy                             noarch                          2.34-481.1.el9_6                           rhel-9-ap
 perl-Math-BigInt                           noarch                          1:1.9998.18-460.el9                        rhel-9-ap
 perl-Math-Complex                          noarch                          1.59-481.1.el9_6                           rhel-9-ap
Installing weak dependencies:
 mariadb-backup                             x86_64                          3:10.5.29-3.el9_7                          rhel-9-ap
 mariadb-gssapi-server                      x86_64                          3:10.5.29-3.el9_7                          rhel-9-ap
 mariadb-server-utils                       x86_64                          3:10.5.29-3.el9_7                          rhel-9-ap

Transaction Summary
================================================================================================================================
Install  13 Packages

Total download size: 18 M
Installed size: 94 M
Downloading Packages:
(1/13): perl-DynaLoader-1.47-481.1.el9_6.x86_64.rpm
(2/13): perl-File-Copy-2.34-481.1.el9_6.noarch.rpm
(3/13): perl-Math-BigInt-1.9998.18-460.el9.noarch.rpm
```

```
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.se
[ec2-user@ip-10-0-0-41 ~]$ sudo mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix socket ensures that nobody
```

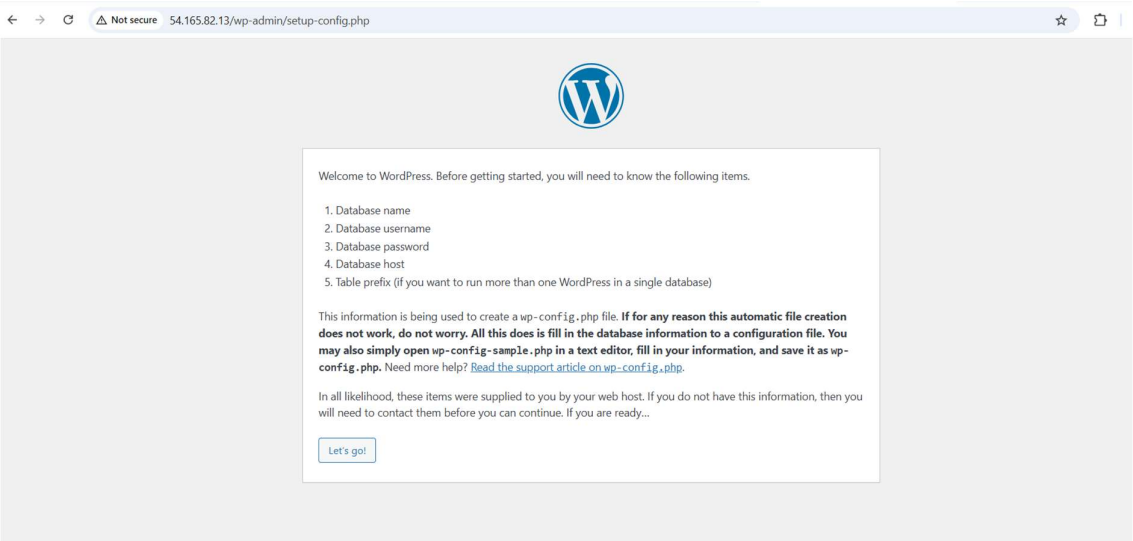
```
[ec2-user@ip-10-0-0-41 ~]$ sudo mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 11
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE wordpress;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> CREATE USER 'wpuser'@'%' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON wordpress.* TO 'wpuser'@'%';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
Bye
```

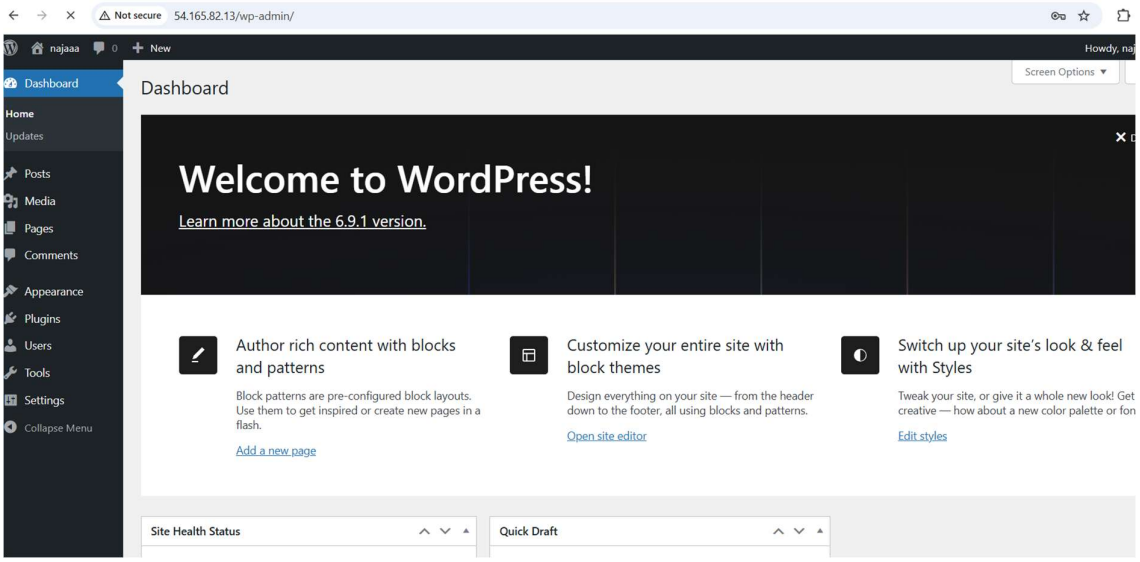
Step 6: Configured WordPress conf file on public subnet

Sudo vi /var/www/html/wp-config.php

```
// ** Database settings - You can get this info from your web host ** //
/** The name of the database for WordPress */
define( 'DB_NAME', 'wordpress' );

/** Database username */
define( 'DB_USER', 'wpuser' );

/** Database password */
define( 'DB_PASSWORD', 'password123' );

/** Database hostname */
define( 'DB_HOST', '10.0.0.41' );

/** Database charset to use in creating database tables. */
define( 'DB_CHARSET', 'utf8mb4' );

/** The database collate type. Don't change this if in doubt. */
define( 'DB_COLLATE', '' );

/**#@+
 * Authentication unique keys and salts.
 *
 * Change these to different unique phrases! You can generate these using
 * the {@link https://api.wordpress.org/secret-key/1.1/salt/ WordPress.org secret-key service}.
 *
 * You can change these at any point in time to invalidate all existing cookies.
```

Step 7: Accessed the WordPress website




Not secure
54.165.82.13/wp-admin/
najaaa
0
New
Howdy, naj
Dashboard
Home
Updates
Posts
Media
Pages
Comments
Appearance
Plugins
Users
Tools
Settings
Collapse Menu
Dashboard
Screen Options
Welcome to WordPress!
Learn more about the 6.9.1 version.
Author rich content with blocks and patterns
Block patterns are pre-configured block layouts. Use them to get inspired or create new pages in a flash.
Add a new page
Customize your entire site with block themes
Design everything on your site — from the header down to the footer, all using blocks and patterns.
Open site editor
Switch up your site's look & feel with Styles
Tweak your site, or give it a whole new look! Get creative — how about a new color palette or fon
Edit styles
Site Health Status
Quick Draft